Exercise1

February 6, 2026

1 Exercise 1

1.1 1.1)

[10]:
```
import autograd.numpy as np
from autograd import grad
from autograd import value_and_grad
# gradient descent function - inputs: g (input function),
# alpha (steplength parameter),
# max_its (maximum number of iterations), w (initialization)
def MA_gradient_descent(g,alpha, beta, max_its,w):
    # compute gradient module using autograd
    gradient = grad(g)

    # run the gradient descent loop
    weight_history = [w]      # container for weight history
    cost_history = [g(w)]     # container for corresponding cost function history
    directions = [-gradient(w)] # container for directions
    for k in range(2, max_its):
        # evaluate the gradient, store current weights and cost function value
        gradient = value_and_grad(g)
        cost_eval,grad_eval = gradient(w)
        dk_1 = beta*directions[k-2] + (1-beta)*(-grad_eval)
        directions.append(dk_1)
        # take gradient descent step
        w = w + alpha*dk_1
        # record weight and cost
        weight_history.append(w)
        cost_history.append(cost_eval)
    return weight_history, cost_history
```

1.2 1.2)

[11]:
```
# defining g
C = np.array([[0.5,0],[0,9.75]])
g = lambda w: np.dot(np.dot(w.T,C),w)

# parameters
```

```
alpha = 0.1
betas = [0, 0.1, 0.7]
w0 = np.array([10.0, 1.0])
max_its = 25

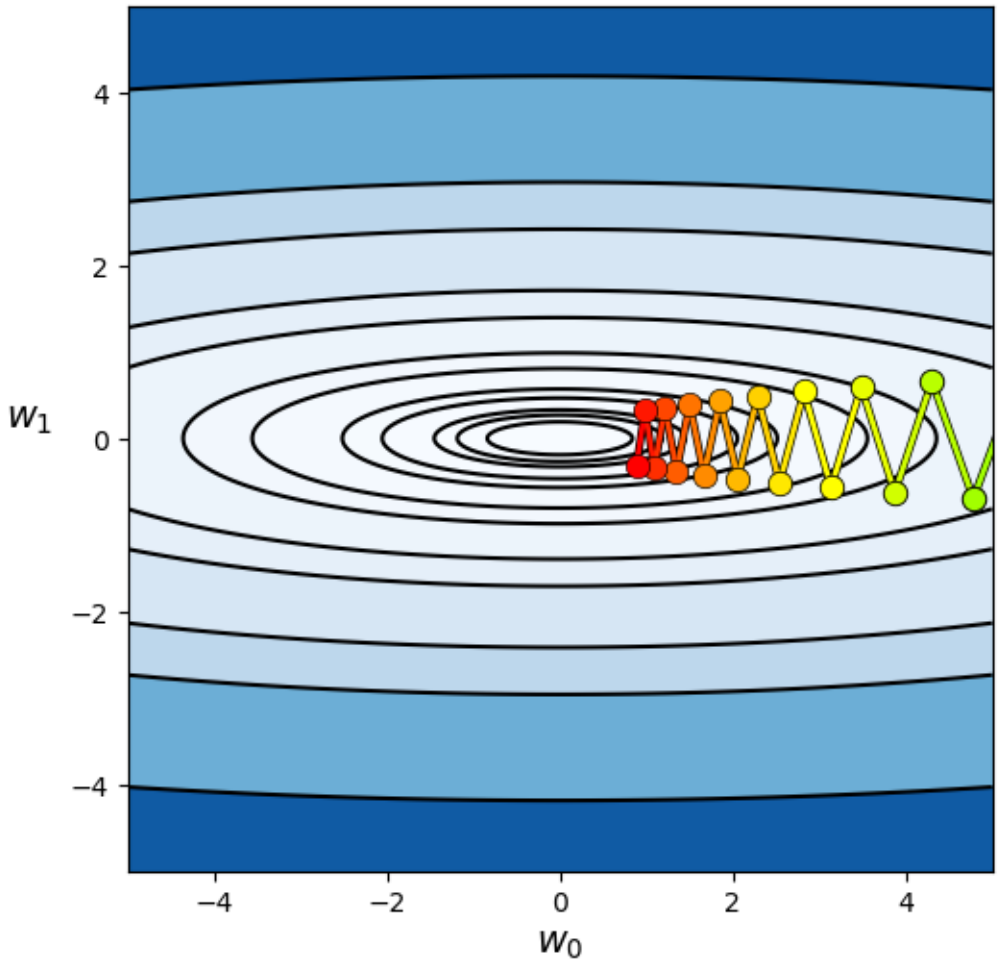
# cost histories
weight_history_1,cost_history_1 = MA_gradient_descent(g,alpha, betas[0],␣
 ↪max_its,w0)
weight_history_2,cost_history_2 = MA_gradient_descent(g,alpha, betas[1],␣
 ↪max_its,w0)
weight_history_3,cost_history_3 = MA_gradient_descent(g,alpha, betas[2],␣
 ↪max_its,w0)
```

[12]:
```
# contour plots of the function g with overlayed w_history to visualize the␣
 ↪paths taken by the algorithms for each beta
from MLRPlotting import static_plotter
# import custom plotting tools
static_plotter = static_plotter.Visualizer()
static_plotter.two_input_contour_plot(g,weight_history_1,xmin = -5,xmax =␣
 ↪5,ymin = -5,ymax = 5,num_contours = 30,show_original = False)
static_plotter.two_input_contour_plot(g,weight_history_2,xmin = -5,xmax =␣
 ↪5,ymin = -5,ymax = 5,num_contours = 30,show_original = False)
static_plotter.two_input_contour_plot(g,weight_history_3,xmin = -5,xmax =␣
 ↪5,ymin = -5,ymax = 5,num_contours = 30,show_original = False)
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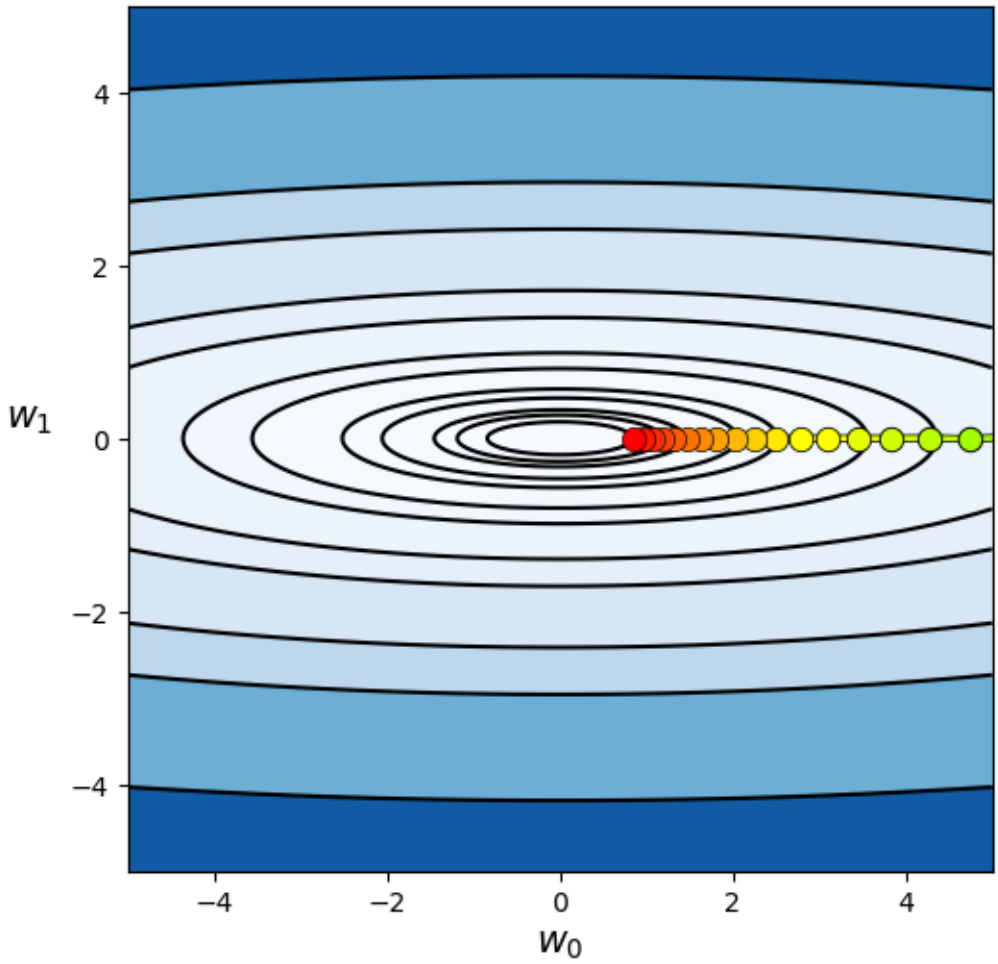
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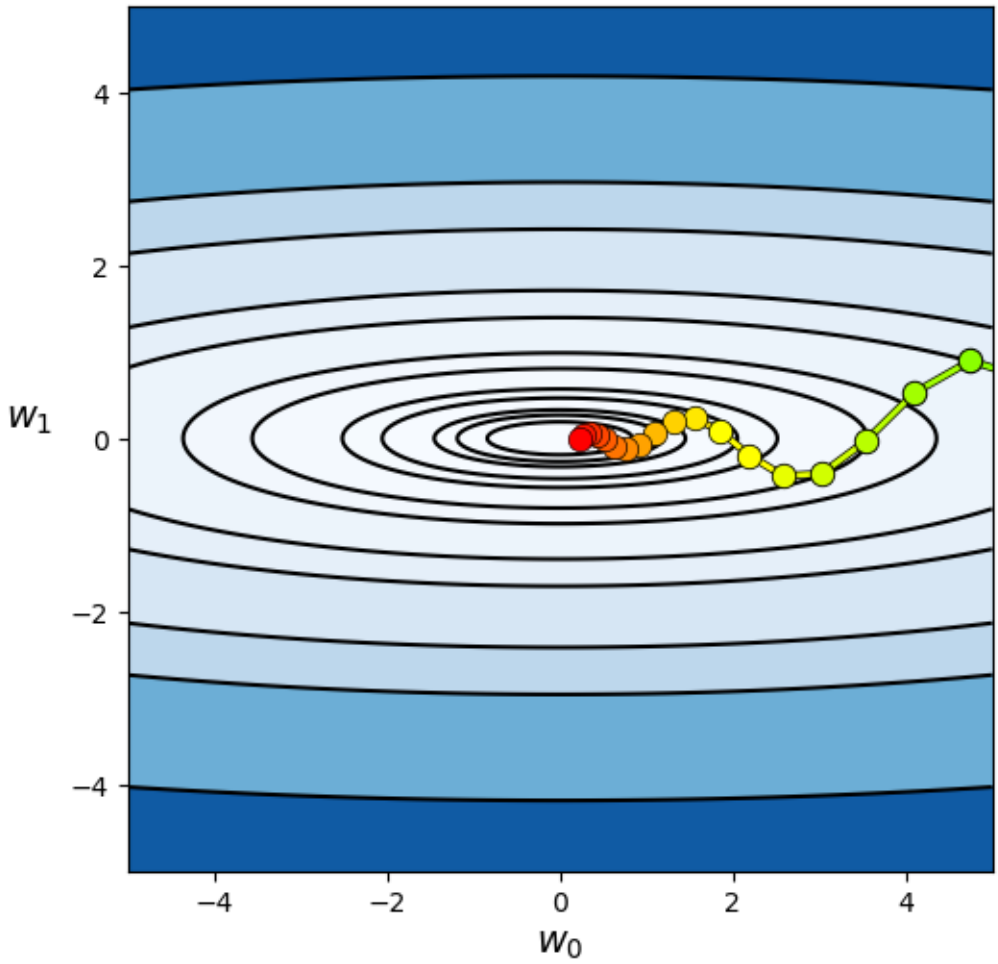
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[13]:
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# plot cost histories
static_plotter.plot_cost_histories([cost_history_1, cost_history_2,␣
↪cost_history_3],start=0,points=True,labels=[r'Gradient descent Beta = 0',␣
↪r'Gradient descent Beta = 0.1', r'Gradient descent Beta = 0.7'])
```

